

of resin and then hardened into the desired shape. In SL methods, after each layer is printed, the object has to be lifted a microscopic level off the build plate to add another layer. This makes the process slow.

On the other hand, Carbon uses a clear, contact lens-like window in the resin pool, which is permeable to light and oxygen. According to the company, this creates a carefully choreographed balance between the light, which hardens the resin, and oxygen, which inhibits hardening. As a result, the object is created continually, without the need to add layers. This also makes it possible to create truly isotropic objects, which have same characteristics on all three axes. Adidas has tied up with Carbon to produce lightweight and durable midsoles for its 4D line of athletic sneakers.

In June, the Korea Electrotechnology Research Institute (KERI) announced the development of a 3D printing technology capable of printing highly-conductive multiwall carbon nanotubes (MWNTs), which will be very useful in wearables, flexible displays and other such applications. The meniscus-guided 3D printing of highly-conductive MWNT microarchitectures exploits rapid solidification of a fluid ink meniscus formed by pulling a micro-nozzle. The team has designed a polyvinylpyrrolidone-wrapped MWNT ink with uniform dispersion in order to achieve smooth 3D printing without any nozzle clogging. This technique gives scientists a lot of control over the printing process, and can be used to print a variety of microstructures with a high concentration of MWNTs.

Optomec, another global leader in 3D printers, has two unique technologies up its sleeve—Aerosol Jet and LENS. Aerosol Jet is used to 3D print electronics at the micron level, while LENS is a metal printing solution that can be added to computer numerically controlled (CNC)

machining systems. Aerosol Jet is an additive manufacturing process that works with a variety of dielectric, conductive, semiconductor and biological inks. It uses aerodynamic focusing to accurately deposit electronic inks onto substrates, printing features ranging from 10 microns to millimetres.

The robotics industry is also an avid user of 3D printing technology. Nanoracing is a two-year-old start-up that makes mini racing drones. Their drones are 3D printed, with absolutely no screws or joints. All the electronic parts fit perfectly into a cavity in the drone.

In an interview with Sculpteo, a company that 3D prints Nanoracing drones, one of the co-founders of Nanoracing explained that 3D printing is an outstanding tool for prototyping and short-run production. "That's what we offered our first clients: an early bird short run of 320 products. And for us, 3D printing allowed us to do the tests, the crash tests, and adapt the product. All of this while avoiding the costs and delays that go along with injection moulding: rheology tools, moulds, injection, and the validations between each step."

Another robotic venture worth a look is InMoov—an open-source, life-size robot made with 3D printed components and controlled by Arduino microcontrollers. Although it is now around five years old, it continues to excite the community. It's amazing because you can download the designs and print your own robot too if you have access to a 3D printer!

Indeed, 3D printing opens up a world of opportunities, for everybody and every industry. In India, it might still be too costly for you to own your own 3D printer. However, costs are going down. There are 3D printers like the da Vinci Mini that are priced around ₹ 40,000, and organisations can invest in these for learning and innovation. **EFY**



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Meco Meters Private Limited
 Plot No. EL-60, MIDC Electronic Zone,
 TTC Industrial Area, Mahape,
 Navi Mumbai - 400710 (INDIA)
 Tel : 091-22-27673300 (Board), 27673311-16 (Sales)
 Fax : 091-22-27673310, 27673330
 Email : sales@mecoind.com
 Web : www.mecoind.com